

HOME SOC LAB
Security Operations Center

CONSOLIDATED SECURITY INCIDENT &
THREAT ANALYSIS REPORT

CONFIDENTIAL — FOR AUTHORIZED PERSONNEL ONLY

Report Period	February 10, 2026 – April 7, 2026
Environment	Home SOC Lab — Wazuh SIEM v4.7.5
Monitored Assets	SERVER1 (172.20.10.2) — Windows Server 2022 Standard Evaluation
Report Classification	CONFIDENTIAL
Report Date	May 30, 2026
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OVERALL RISK RATING

Active brute-force campaigns and critical CIS benchmark compliance failures detected

1. Executive Summary

This report consolidates findings from three Wazuh SIEM security event exports collected across a home SOC laboratory environment between February 10, 2026 and April 7, 2026. The environment consists of a single monitored endpoint — SERVER1 (Windows Server 2022 Standard, 172.20.10.2) — managed by a Wazuh v4.7.5 server.

During the reporting period, the SIEM detected a total of 3,087+ security events across all three reports. The most significant findings include a confirmed brute-force credential attack campaign generating over 2,600 failed authentication attempts, active exploitation-pattern network behaviour, multiple critical CIS Benchmark compliance failures, and evidence of unauthorized system modification events.

The environment presents a HIGH aggregate risk rating. Immediate remediation actions are required to harden authentication controls, enforce CIS benchmark compliance, and investigate anomalous package installation activity.

Metric	Value	Severity
Total Alerts (Combined)	3,087+	—
Total Alerts (Combined)	3,087+	Informational
Authentication Failures	1,302 – 1,334	CRITICAL
Brute-Force Alerts (L10/L13)	8 confirmed incidents	HIGH
CIS Benchmark Score (SERVER1)	34–37% (Fail < 50%)	CRITICAL
Compliance Drift Events	2 critical policy regressions	HIGH
Host-Based Anomaly Events	16 (rootcheck)	HIGH
Network Port Change Events	14 (netstat changes)	MEDIUM
Successful Authentications	218–220 (within attack window)	REVIEW REQUIRED

2. Environment & Monitoring Infrastructure

2.1 Monitored Assets

Agent ID	Hostname	IP Address	OS	Wazuh Version
005	SERVER1	172.20.10.2	Windows Server 2022 (10.0.20348.587)	v4.7.5
N/A	wazuh-server	(Manager)	Linux (Wazuh Manager)	v4.7.5

2.2 Reporting Windows

Report	Start	End
Report 1	2026-02-10 13:49:04 UTC	2026-02-11 13:49:04 UTC
Report 2	2026-04-06 16:40:59 UTC	2026-04-07 16:40:59 UTC
Report 3	2026-04-06 16:42:24 UTC (Agent-focused)	2026-04-07 16:42:24 UTC

Note: Reports 2 and 3 cover overlapping periods; Report 3 is scoped specifically to Agent 005 (SERVER1), providing a focused view of endpoint-level activity.

3. Threat Analysis & Key Findings

3.1 Brute-Force Credential Attack Campaign [CRITICAL]

The most severe finding across all three reporting windows is an active, sustained brute-force credential attack targeting SERVER1. The following alert chain was consistently observed:

Rule ID	Description	Severity	Level	Count
60104	Windows audit failure event	CRITICAL	5	1,332
60122	Logon failure - Unknown user or bad password	CRITICAL	5	1,302
60204	Multiple Windows logon failures	HIGH	10	24
100002	Windows brute-force attack detected	HIGH	10	4
100002	Windows brute-force attack detected (5 attempts in 60s)	CRITICAL	13	4

MITRE ATT&CK Mapping: T1110.001 — Brute Force: Password Guessing | T1078 — Valid Accounts (post-success)

The 218+ successful logon events occurring in the same window as 1,300+ failures raises the possibility that some authentication attempts succeeded. This warrants immediate account review and session audit.

3.2 Authentication Success During Attack Window [HIGH]

218 successful Windows logon events (Rule 60106) and 2 workstation logon successes (Rule 60118) were recorded within the same timeframe as the brute-force campaign. It cannot be confirmed without individual log correlation whether these successes represent legitimate administrator activity or attacker-achieved access. This is the single highest-priority investigative action item.

3.3 CIS Benchmark Compliance Failures [CRITICAL]

SERVER1 scored between 34% and 37% on the CIS Microsoft Windows Server 2022 Benchmark v1.0.0 — well below the minimum passing threshold of 50%. Two critical policy regressions were recorded as status changes from 'passed' to 'failed':

Rule ID	CIS Control	Level	Status
19011	Guest account status changed from Disabled to Enabled (regression)	9	FAIL

19011	Password complexity requirements disabled (regression from passed)	9	FAIL
19007	Account lockout threshold not set to 5 or fewer attempts	7	FAIL
19007	Administrator account not disabled (MS only)	7	FAIL
19007	Allow Remote Shell Access is not disabled	7	FAIL
19007	WinRM remote management allowed (should be disabled)	7	FAIL

The guest account enablement combined with an absence of password complexity enforcement and no account lockout policy creates a trivially exploitable attack surface that directly enabled the observed brute-force campaign.

3.4 Host-Based Anomaly & Rootcheck Detections [HIGH]

16 host-based anomaly detection events (Rule 510 — rootcheck) were flagged across the April 7 reporting window, along with 14 network port state change events (Rule 533 — netstat changes). These events indicate potential unauthorized modification of listening services or running processes.

MITRE ATT&CK Mapping: T1543 — Create or Modify System Process | T1049 — System Network Connections Discovery

3.5 Unauthorized Package Installation Activity [HIGH]

Report 3 (April period) recorded 177 half-configured Debian package events (Rule 2904, Level 7) and 109 successful package installations (Rule 2902, Level 7). This is atypical activity for a Windows Server host unless the Wazuh manager (Linux) is the source. However, the volume and frequency require investigation to rule out malicious package deployment.

MITRE ATT&CK Mapping: T1072 — Software Deployment Tools | T1059 — Command and Scripting Interpreter

3.6 Privilege Escalation Indicators [MEDIUM]

19–40 successful sudo-to-root executions (Rule 5402, Level 3) were recorded across reporting windows, including one first-time sudo execution (Rule 5403, Level 4). While these originate from the Wazuh manager host, the volume and presence of a new user/group creation event (Rules 5901/5902, Level 8) warrants review.

MITRE ATT&CK Mapping: T1548.003 — Abuse Elevation Control Mechanism: Sudo and Sudo Caching

3.7 System Integrity & Stability Events [MEDIUM]

The following system-level events were observed, indicating both instability and potential configuration weaknesses:

Rule ID	Description	Level	Count
61102	Windows System error event	5	27
61110	Multiple System error events (aggregated)	10	3
60730	Inconsistent system shutdown detected	5	1
60602	Windows application error event	9	1

4. Alert Volume & Group Distribution

4.1 Alert Group Summary (Consolidated)

Alert Group	Alert Count	Significance
windows	2,930	Primary Windows security channel
windows_security	2,878	Security event log — direct attack evidence
authentication_failed	1,302	Credential attack volume
authentication_success	220	Requires correlation with failure window
brute_force	8	Confirmed brute-force incidents
windows_system	30	System-level errors and warnings
system_error	28	Repeated system instability
authentication_failures	24	Aggregate failure groupings
sca	5	Security Configuration Assessment events

4.2 PCI DSS Compliance Mapping

The Wazuh reports map triggered alerts to PCI DSS requirements. The top requirements triggered are:

PCI DSS Req.	Description & Implication
10.2.5	Use of identification and authentication mechanisms — directly triggered by 1,302+ logon failures and brute-force alerts
10.2.4	Invalid logical access attempts — maps to audit failure and logon failure events
1.x	Network access controls — triggered by netstat port change events (Rule 533)
2.2	System configuration standards — maps directly to CIS Benchmark failures

5. MITRE ATT&CK Framework Mapping

The following tactics and techniques were identified based on alert correlation across all three reporting periods:

Tactic	Technique	Evidence	Severity
Credential Access	T1110.001 — Password Guessing	1,302 logon failures + 8 brute-force alerts	CRITICAL
Initial Access	T1078 — Valid Accounts	218+ successful authentications during attack window	
Privilege Escalation	T1548.003 — Sudo and Sudo Caching	19–40 sudo-to-root executions; new user/group created	HIGH
Defense Evasion	T1562 — Impair Defenses	PAM misconfiguration event (Rule 5553)	HIGH
Discovery	T1049 — System Network Connections	14 netstat port change events (Rule 533)	MEDIUM
Persistence	T1543 — Modify System Process	16 rootcheck anomaly events (Rule 510)	HIGH
Execution	T1072 — Software Deployment Tools	177 dpkg half-configured + 109 installations	HIGH

6. Recommendations & Remediation Actions

The following remediation actions are prioritized by severity and should be addressed in sequence. Critical items require immediate attention within 24–48 hours.

6.1 CRITICAL — Immediate Actions (0–48 Hours)

#	Action	Target	Control
1	Audit all successful authentications between 12:00–16:00 on April 7, 2026. Identify if any logins during the brute-force window were unauthorized.	SERVER1	T1078 Mitigation
2	Re-enable password complexity requirements on SERVER1 via Group Policy (secpol.msc). Ensure minimum 12-character, multi-class passwords.	SERVER1 GPO	CIS 1.1.5
3	Disable the Guest account on SERVER1 immediately. Confirm via: net user guest /active:no	SERVER1	CIS 2.3.1.2
4	Configure Account Lockout Policy: set threshold to 5 failed attempts, lockout duration 30 minutes, reset counter after 15 minutes.	SERVER1 GPO	CIS 1.2.x
5	Rotate all local and service account credentials on SERVER1. Assume possible compromise until authentication audit is complete.	All accounts	T1110 Mitigation

6.2 HIGH — Short-Term Actions (48 Hours – 1 Week)

#	Action	Target	Control
6	Review and audit the 14 netstat port change events. Identify any unexpected listening services and disable them.	SERVER1	CIS 9.x
7	Investigate 16 rootcheck anomaly events. Run a full integrity check and compare against known-good baseline.	SERVER1	T1543 Mitigation
8	Audit all dpkg install/half-configure events on Wazuh manager. Validate packages against intended software inventory.	wazuh-server	T1072 Mitigation
9	Disable the built-in Administrator account on SERVER1. Create a named admin account with a strong unique password.	SERVER1	CIS 2.3.1.1
10	Disable Remote Shell Access and WinRM on SERVER1. If remote management is required,	SERVER1 GPO	CIS 18.x

	enforce certificate-based WinRM with host restrictions.		
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6.3 MEDIUM — Ongoing Hardening (1–4 Weeks)

- Target CIS Benchmark score above 80% on SERVER1. Prioritize all Level 7+ failed controls identified in SCA reports.
- Implement Wazuh active response rules to automatically block source IPs after 10 failed authentication attempts within 60 seconds.
- Enable audit logging for Authentication Policy Change and Logon/Logoff events per CIS recommendations.
- Investigate inconsistent shutdown event (Rule 60730). Review Windows Event ID 6006/6008 logs to determine if unexpected reboots occurred.
- Resolve the PAM misconfiguration alert (Rule 5553) on the Wazuh manager host.
- Implement a formal change management process for all package installations on the Wazuh server.

7. Risk Register

Risk	Likelihood	Impact	Rating	Owner
Unauthorized access via brute-force (credentials compromised)	HIGH	CRITICAL	CRITICAL	SOC Analyst
Guest account exploitation (enabled with no lockout policy)	HIGH	CRITICAL	CRITICAL	SOC Analyst
Persistence via rootcheck-detected process modification	MEDIUM	HIGH	HIGH	SOC Analyst
Unauthorized software deployment on Wazuh server	MEDIUM	HIGH	HIGH	SOC Analyst
Privilege escalation via uncontrolled sudo usage	MEDIUM	HIGH	HIGH	SOC Analyst
CIS benchmark non-compliance (score < 50%)	HIGH	HIGH	HIGH	SOC Analyst

8. Conclusion

This consolidated analysis of the home SOC lab environment reveals a high-risk security posture driven primarily by an active, sustained brute-force credential attack campaign and critical deficiencies in the baseline security configuration of SERVER1. The combination of no account lockout policy, an enabled Guest account, disabled password complexity requirements, and a CIS Benchmark score below 35% represents a configuration that would not survive even a moderately skilled attacker.

The 218+ authentication successes occurring within the same timeframe as 1,300+ failures is the single most urgent item requiring investigation, as it may indicate a successful compromise. Until this is ruled out, SERVER1 should be treated as potentially compromised.

Secondary concerns — the host-based rootcheck anomalies, unexpected network port changes, and high-volume package installation activity — require prompt investigation to determine whether they represent residual effects of the brute-force campaign, deliberate persistence mechanisms, or legitimate administrative activity.

Systematic remediation following the priority order in Section 6 will significantly reduce the environment's attack surface. A follow-up security assessment should be conducted 30 days after remediation to validate control effectiveness.

Overall Environment Risk Rating	HIGH — Immediate Action Required
Next Review Date	30 days post-remediation
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Report Date	May 30, 2026